Do LLMs Know When They're Making Things Up?

Token probabilities, SelfCheckGPT and a stacked detector on HaluEval QA

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Abstract

This report tests three hallucination detectors on 200 questions from HaluEval QA (400 answers, half hallucinated), using Qwen2.5-3B-Instruct. Mean token log-probability and mean entropy score below random (AUC-PR 0.41 and 0.42). SelfCheckGPT, run with the authors' code, gives 0.65 for the NLI variant and 0.32 for BERTScore. A logistic regression over all four features reaches 0.951 on a held-out 20%, but answer length alone reaches 0.965. Hallucinated answers in HaluEval are about four times longer than right ones, and most of each detector's result follows from that. With length removed, the combined features still score 0.64 AUC-PR.

1 Introduction

The question behind this project is whether a language model's own signals can tell when its answer is made up. Task 1 uses token probabilities. Task 2 reproduces SelfCheckGPT [1], which checks whether the model agrees with itself across several samples. Task 3 combines both in one trained classifier.

The assigned dataset is the QA part of HaluEval [2]. Each question (from HotpotQA) comes with one right answer and one hallucinated answer written by ChatGPT. The two answers look very different: right answers are short spans like *"yes"* or *"Lanoxin"*, while hallucinated answers are usually full sentences that repeat the question. This difference turns out to explain most of the results.

2 Data and setup

200 of the 10,000 HaluEval QA pairs were sampled with `random.Random(42)`. Each pair gives two items: the right answer (label 0) and the hallucinated answer (label 1). The classes are balanced, so random guessing scores AUC-PR = 0.5. A sample of 200 was used because Task 2 needs 20 generations per question on one GPU.

All tasks use `Qwen/Qwen2.5-3B-Instruct` [3] in float16 on an RTX 5060 Ti (16 GB). The prompt is closed-book: a system message (*"Answer the question concisely and factually."*) and the question. HaluEval's `knowledge` paragraph is not given to the model, and nothing is fine-tuned.

	Median tokens	Mean tokens	Range	Example
Right answers (label 0)	3.0	3.7	1–13	*no*
Hallucinated answers (label 1)	14.5	16.1	2–46	*Yes, both documentaries...*

Table 1: Answer length in tokens. The hallucinated answer is longer in 194 of 200 pairs, and ranking by length alone gives AUC-PR 0.972.

Table 1 shows the length gap. Because of it, answer length is reported as a reference next to every detector below.

3 Task 1: Probability baseline

3.1 Method

For each item, the question prompt x and the answer $a = (a_1, \ldots, a_T)$ go through the model in one forward pass. At each answer position, the log-probability of the actual token and the entropy of the next-token distribution are recorded:

$$\bar{\ell}(a) = \frac{1}{T}\sum_{t=1}^{T} \log p_\theta(a_t \mid x, a_{<t}), \qquad \bar{H}(a) = \frac{1}{T}\sum_{t=1}^{T} H(p_\theta(\cdot \mid x, a_{<t}))\,. \tag{1}$$

The hallucination scores are $-\bar{\ell}$ and $\bar{H}$, as in the specification. The minimum token log-probability and maximum token entropy were also kept for comparison.

3.2 Results

Score (higher = more suspicious)	AUC-PR	AUROC
−mean log-prob (spec)	0.406	0.381
mean entropy (spec)	0.417	0.419
−min token log-prob	0.660	0.697
max token entropy	0.610	0.643
Random	0.500	0.500
Answer length only	0.972	0.971

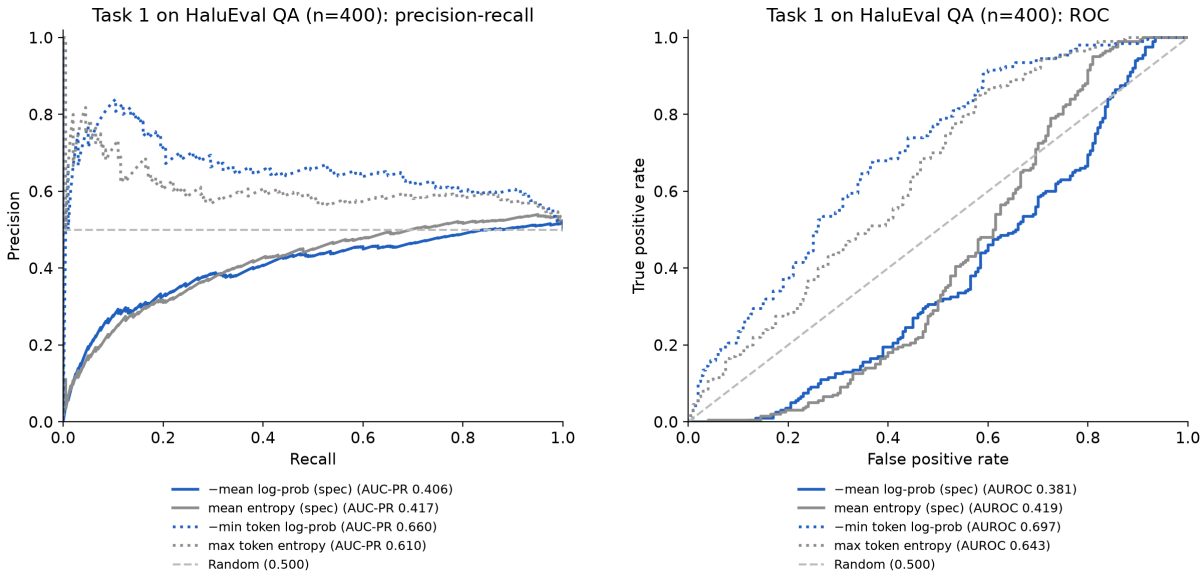
Table 2: Task 1 on all 400 items.

Both specified scores are below random (Table 2, Figure 1). In 63% of pairs the model gives the hallucinated answer a higher mean log-probability than the right one. Median mean log-probability is −3.62 for right answers and −2.49 for hallucinated ones.

The reason is easy to see in a single pair:

> **Pair 1290: the model knows the answer, but log-prob prefers the hallucination**
> *Question:* Do the documentaries Traceroute and War/Dance have the same director?
> *Right:* no — mean log-prob −16.30 (1 token)
> *Hallucinated:* Yes, both documentaries were directed by Johannes Grenzfurthner. — mean log-prob −3.48 (14 tokens)
> *Qwen samples:* "No, Traceroute and War/Dance do not have the same director…" (all 20 samples start with "No")

The model clearly knows the answer, but it would write *"No, …"* as a full sentence. A bare lowercase *"no"* is unlikely as a first token, and with only one token nothing averages it out. The hallucinated answer starts by repeating the question, so most of its tokens are easy to predict. Mean log-probability here measures style more than truth. The worst-token scores do better (0.66) because an invented name is often a single surprising token, though they are also helped by length, since longer answers have more chances to contain a rare token.



(a) Precision–recall **(b)** ROC

Figure 1: Task 1 curves. The two specified scores (solid) fall below the random line; the worst-token scores (dotted) are above it.

4 Task 2: Reproducing SelfCheckGPT

4.1 Method

HaluEval has no pre-generated samples, so $N = 20$ answers per question were generated with the same model and prompt as Task 1 (temperature 1.0, top-$p = 1$, no top-k, at most 48 new tokens, seed 42 + pair_id). Both answers of a pair are scored against the same samples.

Scoring uses the authors' package `selfcheckgpt` 0.1.7. Each answer is passed as one sentence, with the 20 samples as `sampled_passages`:

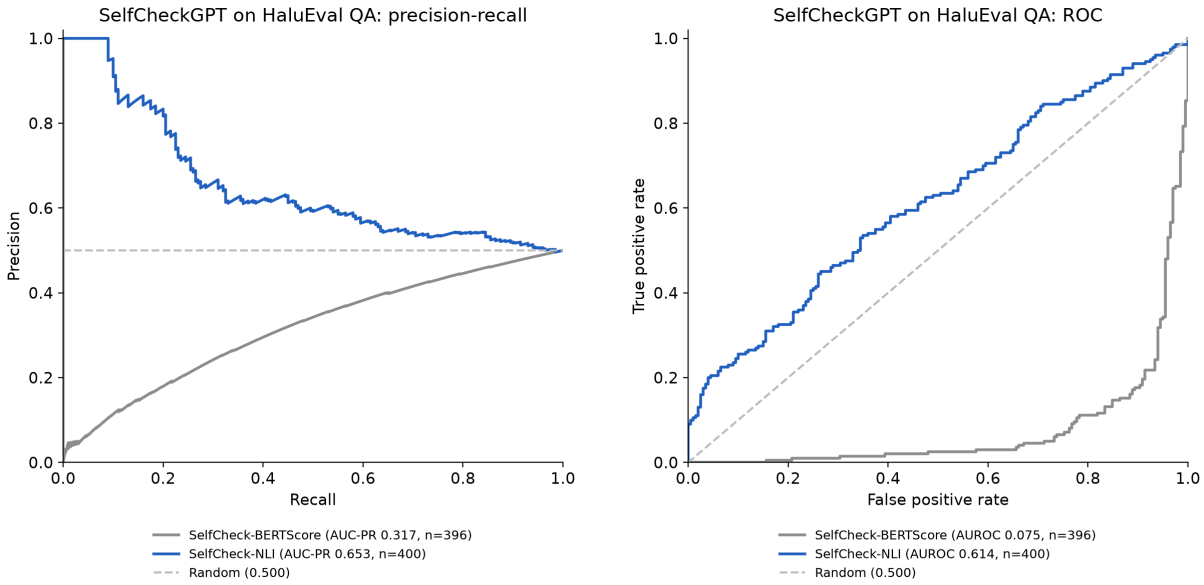
$$S_{\text{BERT}}(r) = 1 - \frac{1}{N}\sum_{n=1}^{N} \max_{k} \text{ BERTScore-F1}(r,\, s_k^n), \tag{2}$$

$$S_{\text{NLI}}(r) = \frac{1}{N}\sum_{n=1}^{N} P(\text{contradiction} \mid r,\, S^n), \tag{3}$$

where s_k^n are the sentences of sample S^n. BERTScore [4] uses `roberta-large` [5] with baseline rescaling. The NLI model is DeBERTa-v3-large [6] trained on MNLI [7].

Deviations.

- `microsoft/deberta-v3-large-mnli`, named in the specification, does not exist on the Hugging Face Hub. `potsawee/deberta-v3-large-mnli` is used instead; it is the checkpoint the official code loads.
- `selfcheckgpt` 0.1.7 calls `batch_encode_plus`, which transformers 5 removed. A short shim (`code/task2/compat.py`) restores it. With the shim, the package reproduces the example scores in its README to six decimal places.
- The BERTScore variant ignores sample sentences of three tokens or fewer. For two questions all 20 samples were single words (e.g. "Canadian"), so those 4 items have no BERTScore ($n = 396$).



(a) Precision–recall **(b)** ROC

Figure 2: Task 2 curves. NLI is above random; BERTScore is far below it, i.e. consistent but in the wrong direction.

4.2 Results

	AUC-PR	AUROC	Median score right	Median score halluc.	Pairs ranked correctly
SelfCheck-BERTScore ($n = 396$)	0.317	0.075	0.959	0.571	6%
SelfCheck-NLI ($n = 400$)	0.653	0.614	0.920	0.966	67%
Random	0.500	0.500	—	—	50%

Table 3: Task 2 on all items. "Pairs ranked correctly": share of questions where the hallucinated answer gets the higher score.

SelfCheck-NLI beats random (Table 3, Figure 2), with good precision at the top of the ranking: at 20% recall, 83% of flagged answers are hallucinated. Still, even right answers have a median contradiction score of 0.92.

SelfCheck-BERTScore is inverted. It scores the right answer as more suspicious in 94% of pairs; with the sign flipped it would get 0.918. The cause is again length:

> **Pair 117: why BERTScore runs backwards**
>
> *Question:* Jake David Shapiro was best know for being the screenwriter of a film directed by who?
>
> *Right:* Roger Christian $S_{\text{BERT}} = 1.10$
>
> *Hallucinated:* Jake David Shapiro was best known for being the screenwriter of a film directed by Steven Spielberg. $S_{\text{BERT}} = 0.34$
>
> *Qwen samples:* "Jake David Shapiro was best known for being the screenwriter of the film 'Superman II' which was directed by Richard Donner."

The samples repeat the question, so a hallucinated answer that also repeats it overlaps with them heavily. A two-word answer has low recall against any full sentence, whatever it says.

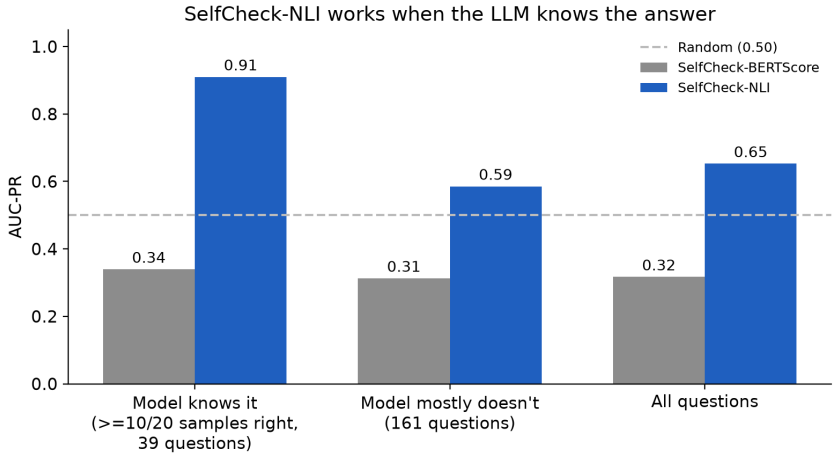


Figure 3: AUC-PR split by whether at least 10 of the 20 samples contain the right answer. With a cut-off of 5 the known-question NLI score is 0.89; with 15 it is 0.93.

4.3 When self-consistency works

SelfCheckGPT assumes the model's samples are mostly right. For each question, the number of samples containing the right answer was counted. Only 41% of questions have even one such sample, and only 20% (39 questions) have at least 10 of 20.

On the 39 questions the model knows, SelfCheck-NLI reaches **0.91**; on the other 161 it falls to 0.59 (Figure 3). Leaving out the 12 yes/no questions among the 39 still gives 0.90. Two failure patterns appear on unknown questions:

> **Pair 1169: the model is consistently wrong**
>
> *Question:* Which film director directed the internet video serial "God, Inc.", Roman Karmen or Francis Stokes?
> *Right:* Francis Stokes $S_{\text{NLI}} = 0.999$
> *Hallucinated:* Roman Karmen directed "God, Inc." with Francis Stokes. $S_{\text{NLI}} = 0.965$
> *Qwen samples:* "Roman Karmen directed the internet video serial 'God, Inc."' (all 20 samples name Roman Karmen)
>
> **Pair 9763: the model does not know**
>
> *Question:* In which city did this world games take place when Squash was added as part of the world games?
> *Right:* Lahti, Finland $S_{\text{NLI}} = 0.999$
> *Hallucinated:* The World Games was never held in Lahti. $S_{\text{NLI}} = 0.990$
> *Qwen samples:* "Toronto"; "Riga, Latvia"; "Doha, Qatar"; "...Kuala Lumpur, Malaysia in 2017."

In pair 1169 the samples agree with each other but not with the truth, so the right answer is flagged. SelfCheckGPT measures agreement with the model, not with facts. In pair 9763 the samples scatter, so both answers are contradicted and nothing separates them.

4.4 Comparison with the paper

The paper reports results on WikiBio, where 73% of sentences are inaccurate, so random already scores 72.96. Table 4 therefore also shows the gain over random.

NLI's gain over random (+15) is close to the paper's (+19.5), and NLI beats BERTScore as in the paper. The other rows do not reproduce. Likely causes:

	Paper (WikiBio, NonFact)		This work (HaluEval QA)	
Method	AUC-PR	vs. random	AUC-PR	vs. random
Random	72.96	—	50.0	—
Avg($-\log p$)	83.21	+10.3	40.6	−9.4
SelfCheck-BERTScore	81.96	+9.0	31.7	−18.3
SelfCheck-NLI	92.50	+19.5	65.3	+15.3
SelfCheck-NLI, known questions	—	—	91.0	+41.0

Table 4: Reproduction vs. the numbers in the official SelfCheckGPT repository (AUC-PR ×100).

1. **Label source.** In WikiBio, the scored text was written by the model being tested. In HaluEval, ChatGPT wrote the hallucinations, so Qwen has no reason to be unsure about them, and they differ in style from the right answers.
2. **Model knowledge.** GPT-3 (175B) on biographies vs. a 3B model on multi-hop questions it mostly cannot answer. On questions it does know, NLI matches the paper.
3. **Scoring unit.** The paper compares sentences with similar sentences; here one-word answers are compared with full-sentence samples, which hurts BERTScore most.
4. **Scale.** 400 items give wide intervals (about ±0.1 for NLI on the test split).

5 Task 3: Stacked ensemble

5.1 Method

The 200 *questions* were split 80/20 with `train_test_split(random_state=42)` (320 training items, 80 test items). Splitting by question keeps both answers of a question in the same part. The four features ($\bar{\ell}$, $\bar{H}$, S_{BERT}, S_{NLI}) are standardised and fed to a logistic regression (L2, $C = 1$). The four missing BERTScore values are filled with the training median.

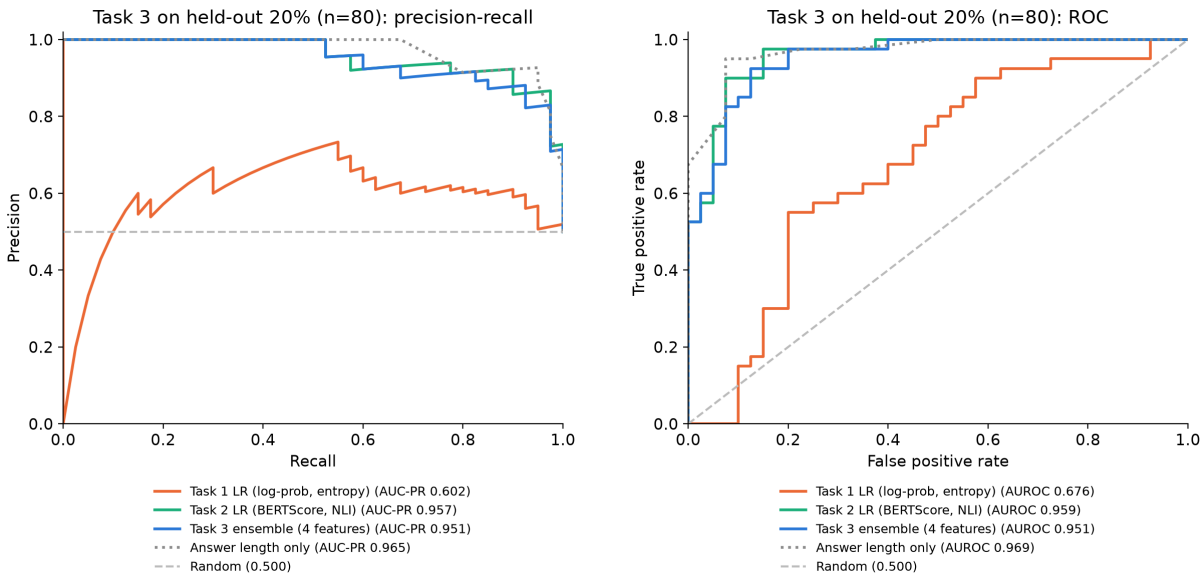
All detectors are evaluated on the same 80 test items. Task 1 and Task 2 are shown both as raw scores and as a logistic regression on their own two features; the second is the fairer comparison. 95% intervals come from 1000 bootstrap resamples of the 40 test questions.

5.2 Results

Detector (held-out 20%, $n = 80$)	AUC-PR	95% CI	AUROC
Task 1: −mean log-prob	0.429	0.40–0.51	0.422
Task 1: mean entropy	0.459	0.42–0.54	0.491
Task 1: LR(log-prob, entropy)	0.602	0.53–0.72	0.676
Task 2: SelfCheck-BERTScore	0.323	0.32–0.34	0.075
Task 2: SelfCheck-NLI	0.651	0.56–0.75	0.602
Task 2: LR(BERTScore, NLI)	0.957	0.91–0.99	0.959
Task 3: LR on all four features	**0.951**	**0.90–0.98**	**0.951**
Task 3+: also min log-prob, max entropy	0.973	0.95–1.00	0.972
Answer length only	0.965	0.94–0.99	0.969
Random	0.500	—	0.500

Table 5: All detectors on the same held-out 80 items.

Task 3 reaches 0.951 (Table 5, Figure 4), far above the specified Task 1 and Task 2 scores. It ranks 38 of 40 test pairs correctly and labels 87.5% of items correctly at a 0.5 threshold. However,



(a) Precision–recall **(b)** ROC

Figure 4: Held-out curves for each task's trained detector, with answer length as a reference.

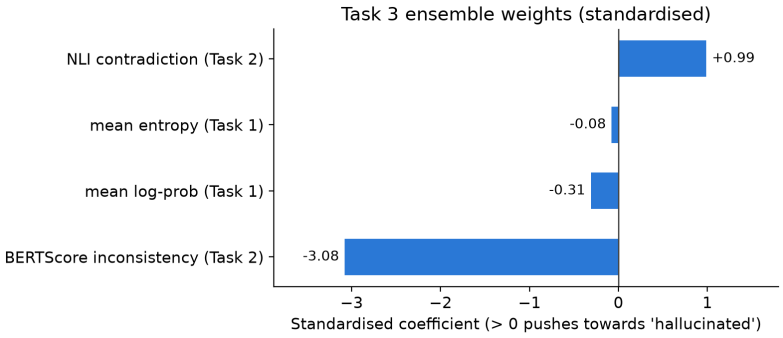


Figure 5: Standardised weights of the Task 3 regression. Positive pushes towards "hallucinated".

it is level with Task 2's own regression (0.957) and just below length alone (0.965); the intervals overlap. Over 100 random splits the picture is the same: LR(Task 1) 0.627 ± 0.047, LR(Task 2) 0.948 ± 0.025, Task 3 0.948 ± 0.025, length 0.972 ± 0.014.

The weights explain this (Figure 5). The largest is BERTScore at -3.08: the classifier uses BERTScore in reverse, as Section 4 suggests. BERTScore correlates with log answer length at $r = -0.70$, so most of the ensemble's signal is length.

5.3 Controlling for length

Two checks, both fitted on training data only:

1. **Length as a feature:** compare LR on log(tokens) alone with LR on log(tokens) plus the four features.
2. **Length removed:** regress each feature on log(tokens), keep the residuals, and train the regression on those.

The features do carry information beyond length (Figure 6, Table 6). Added to length, they raise AUC-PR from 0.972 to 0.987 and help on 95 of 100 splits. With length removed, 0.64

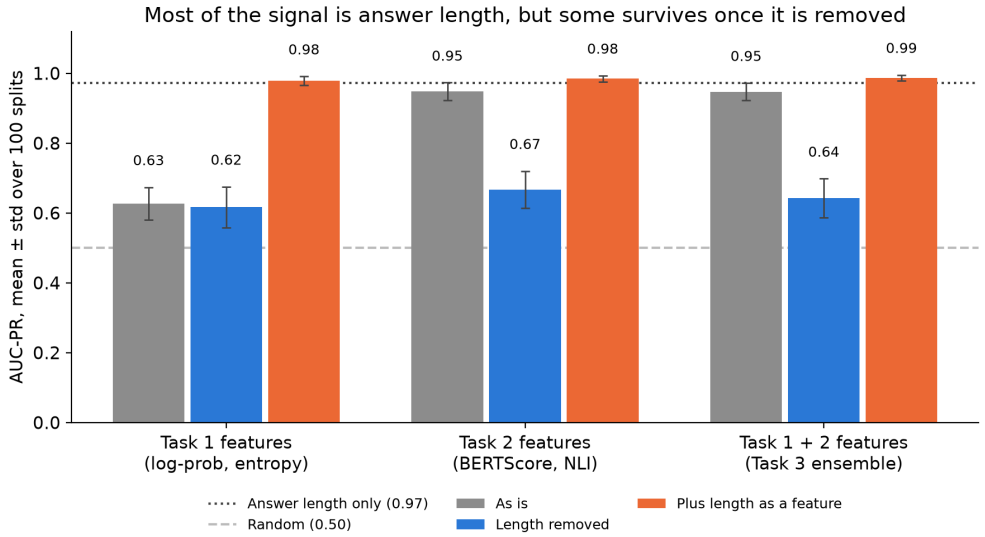


Figure 6: Mean AUC-PR over 100 question-level splits (bars: ±1 sd). Grey: features as they are. Blue: length removed. Orange: length added as a feature.

Detector (100 splits)	AUC-PR	Pairs correct
LR(length)	0.972 ± 0.014	97%
LR(length + Task 1 + Task 2)	0.987 ± 0.008	98%
LR(Task 1, length removed)	0.616 ± 0.059	74%
LR(Task 2, length removed)	0.667 ± 0.053	71%
LR(Task 1 + Task 2, length removed)	0.642 ± 0.056	75%
−mean log-prob alone, length removed	0.614 ± 0.063	75%

Table 6: Length controls, mean ± sd over 100 splits.

AUC-PR remains and three out of four pairs are ranked correctly.

Mean log-probability changes the most. On its own it was below random, but with length removed it ranks 75% of pairs correctly. Length had been pushing it the wrong way. NLI goes the other direction: after the correction it is no better than random (0.44 on the seed-42 split), so part of its raw advantage was length too.

5.4 Error analysis

The ensemble mis-orders 2 of the 40 test pairs. In both, the "hallucinated" answer is actually correct. In pair 6572, *"... is The Chipmunks"* is a paraphrase of *"Alvin and the Chipmunks"*. Pair 3657 is shown below; for this question all 20 Qwen samples say "Canadian", so the model itself is wrong.

Pair 3657: a label problem in HaluEval

Question: ...songs by Heart, a rock band that first found success in Canada and later in the United States and worldwide, are of which nationality?
Right: American — ensemble 0.74
Hallucinated: Heart, the famous rock band, is American. — ensemble 0.62

There are five false negatives at the 0.5 threshold. Three are specific but wrong (e.g. "Cancun, Mexico" instead of Daytona Beach), one is close to a paraphrase, and one makes no checkable claim:

Pair 2060: a vague answer nothing contradicts

Question: Turbidimetric inhibition immunoassay is used in many commercial immunoassays, including this medication sold under which brand name?
Right: Lanoxin ensemble 0.01, $S_{\text{NLI}} = 0.93$
Hallucinated: Turbidimetric inhibition immunoassay is used in many commercial immunoassays, including several medications. ensemble 0.37, $S_{\text{NLI}} = 0.32$

The answer avoids the question instead of inventing a fact, so the samples cannot contradict it. Self-consistency methods cannot catch answers that say nothing.

6 Discussion

What worked. SelfCheck-NLI works when its assumption holds: 0.91 AUC-PR on questions the model knows, with no training and no access to model internals. Splitting by question and reporting a length-only baseline made it clear which results are meaningful.

What did not. The two specified Task 1 scores are below random and SelfCheck-BERTScore is inverted. The Task 3 ensemble looks strong, but mostly because it learns length through BERTScore.

Why. Two reasons. First, the hallucinations were written by a different model in a different style, so they are not tied to Qwen's uncertainty, which all these methods rely on. Second, a 3B model answering multi-hop questions closed-book usually does not know the answer, and self-consistency gives no signal (or the wrong one) in that case.

Limitations. Only 200 of 10,000 questions were used, so intervals are wide. The "known question" split uses string matching and misses paraphrases. The length correction is linear and may not remove all of it. A length-matched test set was not possible: only 28 pairs have answers within five tokens of each other. A cleaner test would score answers that Qwen generated itself, labelled against HotpotQA's gold answers.

7 Conclusion

On HaluEval QA with a 3B open model, token probabilities do not detect hallucinations, and SelfCheckGPT only works on the questions the model can answer. The stacked regression reaches 0.95 AUC-PR, but almost all of that comes from answer length. With length controlled, the combined features rank three out of four pairs correctly, which is the more realistic measure of what they can do.

References

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A Reproducibility

Code, a pinned `requirements.txt` and run instructions are in the submitted archive (`README.md`). With Docker, everything runs with one command, `bash run_docker.sh` (add `-recompute` to redo the GPU steps). Without Docker, it is `bash code/run_all.sh`.

Item	Setting
Data	HaluEval `qa_data.json` (github.com/RUCAIBox/HaluEval), 200 pairs, `random.Random(42)`
LLM	`Qwen/Qwen2.5-3B-Instruct`, float16, closed-book chat prompt
Resampling	$N = 20$, temperature 1.0, top-p 1.0, top-k off, max 48 new tokens, seed $42 +$ pair_id
BERTScore	`roberta-large` via `bert-score` 0.3.13, baseline rescaling
NLI	`potsawee/deberta-v3-large-mnli`
SelfCheckGPT	`selfcheckgpt` 0.1.7 (official code, commit `19b492a`)
Split	question-level 80/20, `train_test_split(random_state=42)`
Classifier	`StandardScaler` + `LogisticRegression(C=1, random_state=42)`
CIs	1000 bootstrap resamples of test questions, `default_rng(42)`
Extra splits	100 more splits for the length check, `random_state` $0, \ldots, 99$
Hardware	NVIDIA RTX 5060 Ti 16 GB; Python 3.12, PyTorch 2.14, transformers 5.21